

Data Flow Diagram & User Stories

Project Design Phase-II

Date	31/07/2026
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Project Name	Intergalactic Space Travel Request System (ISTR)
Maximum Marks	4 Marks

Data Flow Diagram

The diagram below shows the Level 0 data flow for the ISTR system: how a mission request enters the system from the Requester, moves through automated approval routing (Manager, Finance Officer, and — for High/Critical risk missions — Mission Control), is written to and read from the u_space_travel_request table and the audit/SLA log, and results in a launch action and stakeholder notifications.

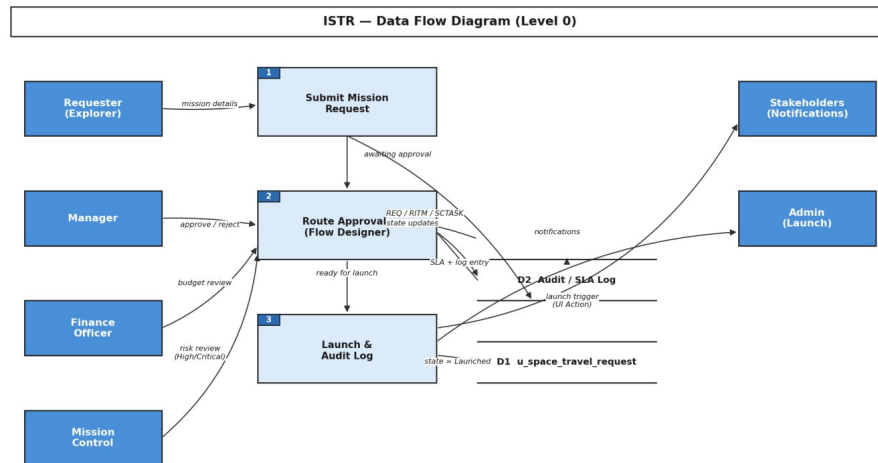


Figure 1 — ISTR Level 0 Data Flow Diagram

Actor–System Context Diagram

Complementing the data flow view, the context diagram below frames ISTR from the outside in: every actor that interacts with the system and the nature of that interaction, independent of the internal processing steps.

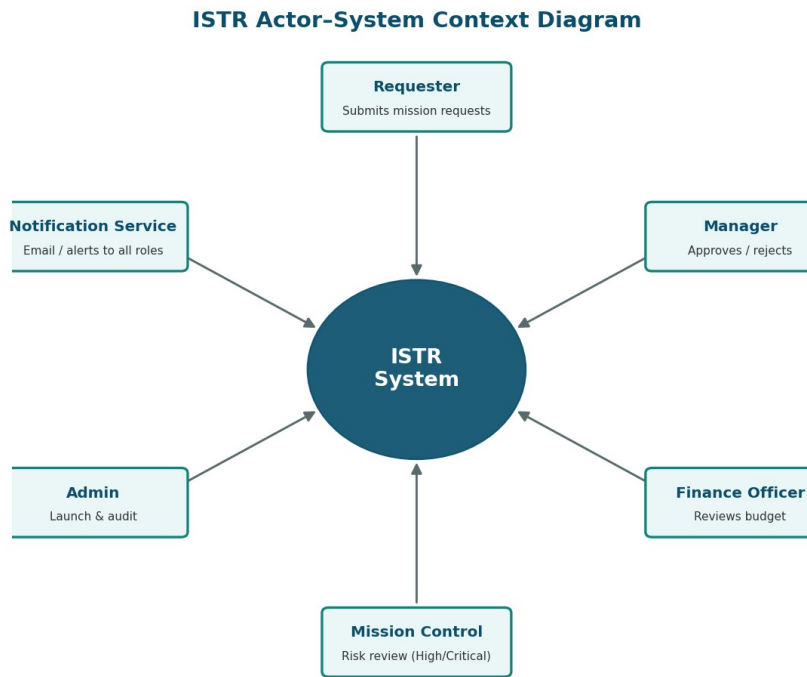


Figure 2 — ISTR Actor-System Context Diagram

User Stories

The table below lists the user stories for the ISTR product, grouped by user type and functional requirement (epic), with acceptance criteria, priority, and target release.

User Type	Functional Requirement (Epic)	USN	User Story / Task	Acceptance Criteria	Priority	Release
Requester (Explorer)	Mission Request Submission	USN-1	As a Requester, I can submit a Mission Request by entering destination, mission type, risk level, cost estimate, and launch date.	A REQ/RITM/SCTASK record is created in u_space_travel_request with state 'Awaiting Manager Approval'.	High	Sprint-1
		USN-2	As a Requester, I can see additional fields appear when I select 'Alien Negotiation' as the mission type.	A Diplomatic Officer field becomes mandatory and visible only for that mission type.	Medium	Sprint-1
		USN-3	As a Requester, I receive an email confirmation once my request is submitted.	Confirmation email is sent with request number and current status.	High	Sprint-1
		USN-4	As a Requester, I can track the live status of my request on a personal dashboard.	Dashboard shows current stage, approver, and SLA countdown.	Medium	Sprint-2
Manager	Approval Routing	USN-5	As a Manager, I am automatically notified when a direct report submits a mission request.	Flow Designer looks up the requester's manager and routes the approval task to them.	High	Sprint-1
		USN-6	As a Manager, I can approve or reject a request with comments.	Approval/rejection updates the state and logs the comment in the audit trail.	High	Sprint-1
		USN-7	As a Manager, I am reminded if I have not actioned a request within the SLA window.	An escalation notification is sent at 80% of SLA duration.	Medium	Sprint-2
Finance Officer	Budget Review	USN-8	As a Finance Officer, I can review and adjust the estimated mission cost.	Cost field is editable only by the space.finance role; changes are logged.	High	Sprint-1
		USN-9	As a Finance Officer, I can approve the budget so the request proceeds to the next stage.	On approval, state moves to 'Awaiting Mission Control' (if High/Critical risk) or 'Ready for Launch'.	High	Sprint-2

User Type	Functional Requirement (Epic)	USN	User Story / Task	Acceptance Criteria	Priority	Release
Mission Control	Risk Assessment	USN-10	As a Mission Control reviewer, I only see requests flagged High or Critical risk in my queue.	ACL and filter restrict the queue to risk-qualified records.	High	Sprint-2
		USN-11	As a Mission Control reviewer, I can request additional information before approving.	Record is returned to the Requester with a comment; state reverts to 'Draft'.	Medium	Sprint-2
Administrator	Launch Execution	USN-12	As an Admin, I can trigger the 'Launch Mission' UI action only when a request is fully approved.	Action is visible only when state = 'Ready for Launch' and role = space.admin.	High	Sprint-3
		USN-13	As an Admin, I can view a complete audit log of every state change and approval.	Audit related list shows timestamp, user, old value, and new value for each change.	Medium	Sprint-3
All Stakeholders	Notifications & Dashboard	USN-14	As a stakeholder, I receive a real-time notification whenever a mission I'm involved in changes state.	Email/Slack-style notification fires on every state transition.	Medium	Sprint-3
		USN-15	As a stakeholder, I can view a role-specific dashboard showing only the missions relevant to me.	Dashboard widgets are filtered by role (Requester, Manager, Finance, Mission Control, Admin).	Low	Sprint-3

User Story Distribution

15 stories are spread across three sprints and three priority levels. Sprint-1 is intentionally the heaviest, front-loading the core submission-to-approval path so later sprints build on a working foundation; High-priority stories dominate the backlog, reflecting that most of ISTR's value sits in the core workflow rather than the peripheral notification/dashboard layer.

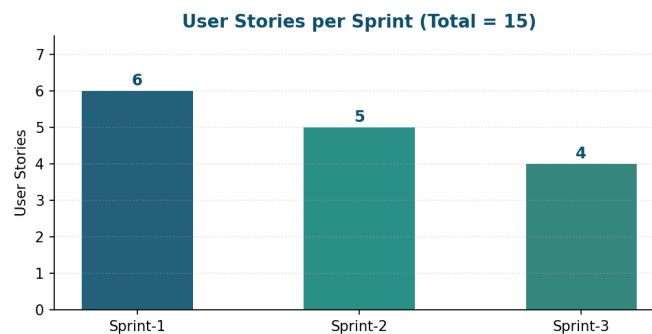


Figure 3 — User Stories per Sprint

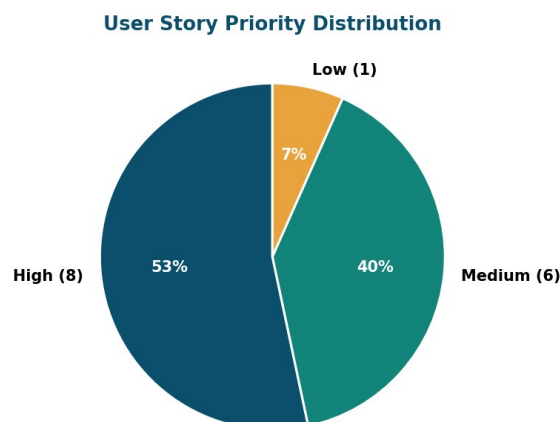


Figure 4 — User Story Priority Distribution